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EUROPE

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From YANG To YAML: How We Tamed the 5G Configuration Beast

Peter Wörndle (Ericsson) & Joel Studler (Swisscom)

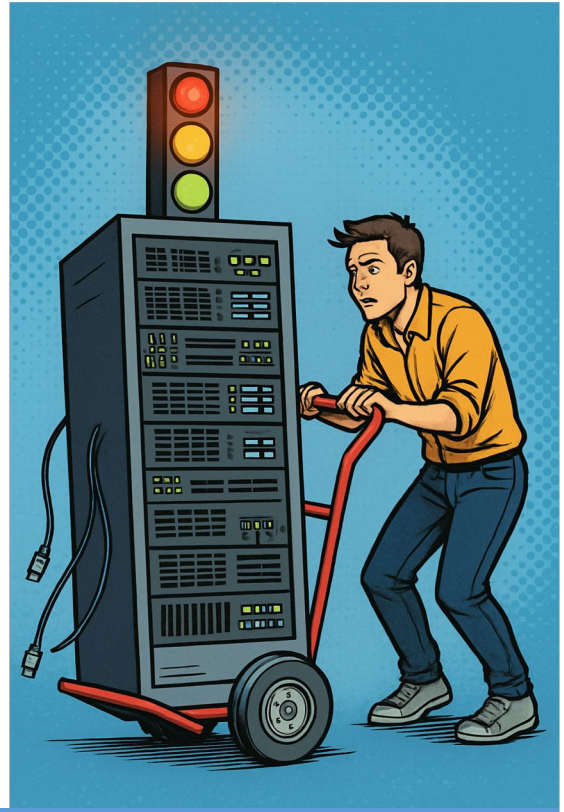
Overview

- A little story about YANG
- A supplier-operator dev hackathon
- kro & SDC
- GitOps & KRM in the telecom domain
- How do we work together
- How do we work with you (the industry)
- Your turn: Q&A

A Little Story About YANG - Chapter 1

Once upon a time …

- Network equipment were large boxes with a status lights on top
- Single Northbound Interface for the box
- Configuration & Observability combined



Networks are still predominantly managed “box-oriented”

A Little Story About YANG - Chapter 2

And there was YANG

- YANG (Modeling) & NETCONF (Protocol)
  - Modeling
    - IETF standardized data models
    - Constraints and conditionals
  - Actions (RPC)
  - Notifications
  - Operational state

```
module acme-system {
    namespace "http://acme.example.com/system";
    prefix "acme";

    import ietf-yang-types {
        prefix "yang";
    }

    include acme-types;

    organization "ACME Inc.";
    contact
        "Joe L. User

         ACME, Inc.
         42 Anywhere Drive
         Nowhere, CA 95134
         USA

         Phone: +1 800 555 0100
         EMail: joe@acme.example.com";

    description
        "The module for entities implementing the ACME protocol.";

    revision "2007-06-09" {
        description "Initial revision.";
    }

    // definitions follow...
}
```

Figure 1: YANG module declaration (REST), RFC6020

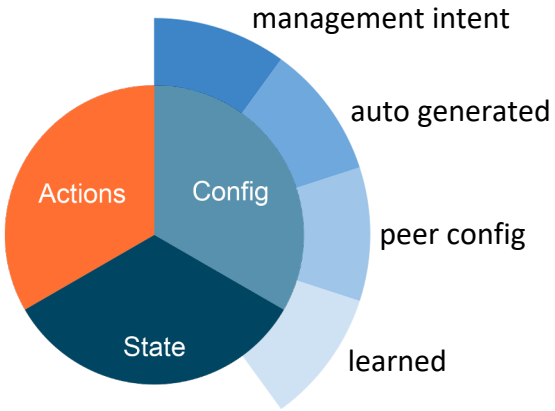
```
<rpc message-id="101"
     xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
  <rock-the-house xmlns="http://example.net/rock">
    <zip-code>27606-0100</zip-code>
  </rock-the-house>
</rpc>

<rpc-reply message-id="101"
           xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
  <ok/>
</rpc-reply>
```

Figure 2: YANG action declaration (XML), RFC6020

A Little Story About YANG - Chapter 3

Challenges today

- Mix of information in a typical model
- Complex & large models
- Not fully declarative (actions, lock/unlock, …)
- Huge install base for both Network Functions (NF) and management systems
- Many NFs are managed as “boxes with status lights”



Types of information in a typical YANG model

A Little Story About YANG - Chapter 4

How to get to declarative CM?

- Revise models
  - Identify declarative aspects
  - Offload “actions” where possible
- Complement with dedicated observability
- Gradually move away from Netconf for CM

Next chapter: PoC & Hackathon …

Vendor-Customer PoC/Hackathon: Overview

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Problem Statement:
Can we use **SDC** and **kro** with **GitOps** to demonstrate benefits of declarative configuration for a **relevant recurring configuration use case** in our 5G Core?

Timeline:
2.5 intense hackathon-style working days

Joint Team:
Very diverse team from both Swisscom and Ericsson

The Team

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Swisscom

Amel Dussier
Integration Engineer

Alessio Diamanti
DevOps Engineer

Alexander North
DevOps Engineer

Ericsson

Ivan Simunic
Solution Architect

Stjepan Sore
Integration Engineer

Jürgen Mollus
Network Program Director

Alexander Langer
Sr Spec Signaling SW

Prashant Pratik
Signaling Developer

Volker Kleinfeld
Signaling System Management

Vera Simon
Signaling Solution Line

Vendor-Customer Hackathon: SDC

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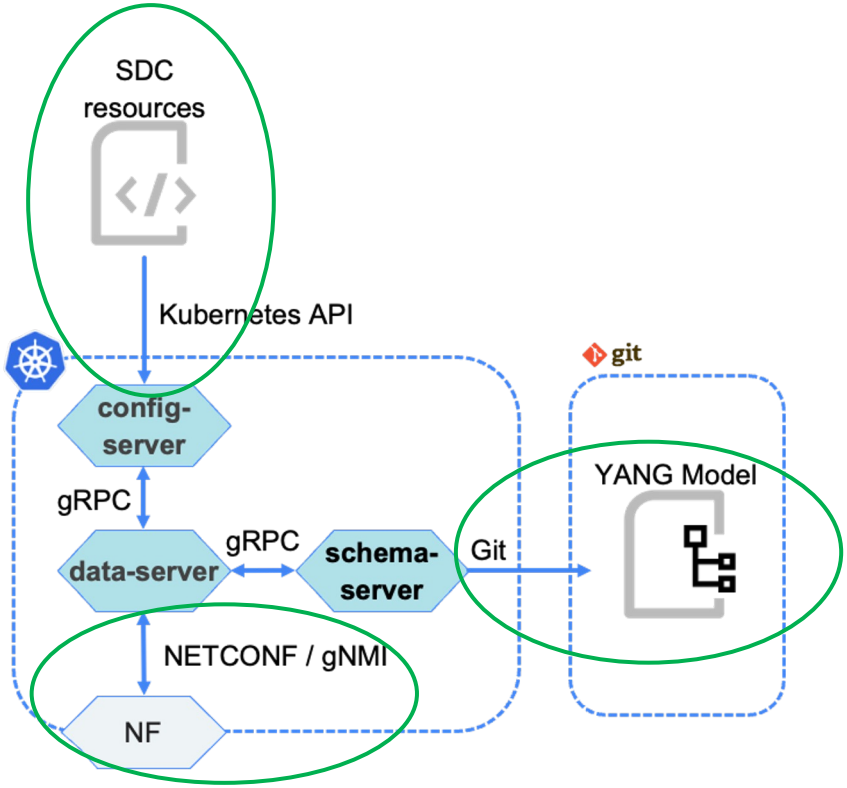
Schema Driven Configuration
https://docs.sdcio.dev

- ✅ Kubernetes-Native (KRM)
- ✅ NETCONF & gNMI
- ✅ YANG Validation
- ✅ Vendor Agnostic
- ✅ CNF, VNF, PNF support
- ✅ Reconciliation

- ❌ Config Abstraction



Vendor-Customer Hackathon: kro

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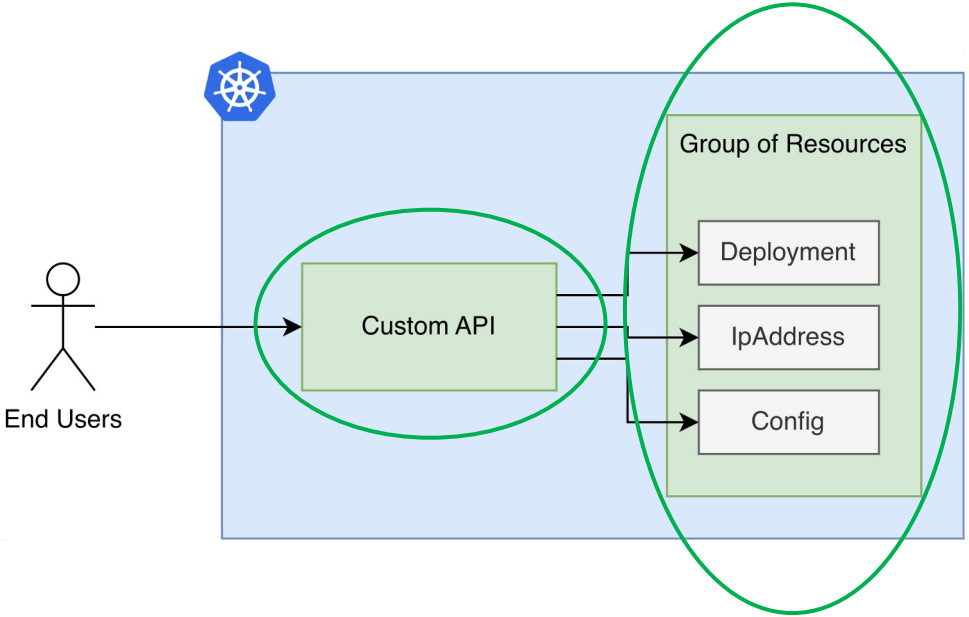
Kube Resource Orchestrator
https://kro.run

- ✅ Kubernetes-Native (KRM)
- ✅ Abstraction Layer
- ✅ Common Expression Language
- ✅ Drift Detection



Vendor-Customer Hackathon: kro+SDC

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```yaml
apiVersion: telco.hack.com/v1
kind: Dnn
spec:
  hostname: "gprs.hack.com"
  ipv4: true
  region: ch-east
  type: MobileInternet
  snssai: [1, 10]
```

reconcile

kro

create

```yaml
apiVersion: config.sdcio.dev/v1alpha1
kind: Config
metadata:
  labels:
    sdcio.dev/targetName: ch-east  # from Dnn.spec.region
    sdcio.dev/targetNamespace: default
spec:
  priority: 10
  config:
  - path: /
    value:
      dnn:
      - name: "gprs.hack.com"  # from Dnn.spec.hostname
        ip: 1.2.3.4  # from IpAddressClaim.status.ip
        snssai: [1, 10]  # from Dnn.spec.snssai
        type: MobileInternet  # from Dnn.spec.type
```

create

```yaml
apiVersion: netbox.dev/v1
kind: IpAddressClaim
spec:
  parentPrefixSelector:
    region: "ch-east"  # from Dnn.spec.region
    family: "IPv4"  # from Dnn.spec.ipv4
status:
  ip: 1.2.3.4
```

reconcile

SDC

NETCONF

CNF

Vendor-Customer Hackathon: kro+SDC

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```yaml
apiVersion: telco.hack.com/v1
kind: Dnn
spec:
  hostname: "gprs.hack.com"
  ipv4: true
  region: ch-east
  type: MobileInternet
  snssai: [1, 10]
```

reconcile

kro

create

create

```yaml
apiVersion: netbox.dev/v1
kind: IpAddressClaim
spec:
  parentPrefixSelector:
    region: "ch-east"  # from Dnn.spec.region
    family: "IPv4"  # from Dnn.spec.ipv4
status:
  ip: 1.2.3.4
```

```yaml
apiVersion: config.sdcio.dev/v1alpha1
kind: Config
metadata:
  labels:
    sdcio.dev/targetName: ch-east  # from Dnn.spec.region
    sdcio.dev/targetNamespace: default
spec:
  priority: 10
  config:
  - path: /
    value:
      dnn:
        name: "gprs.hack.com"  # from Dnn.spec.hostname
        ip: 1.2.3.4  # from IpAddressClaim.status.ip
        snssai: [1, 10]  # from Dnn.spec.snssai
        type: MobileInternet  # from Dnn.spec.type
```

reconcile

SDC

NETCONF

CNF

Vendor-Customer Hackathon: Outcome

Yes, it works.

Yes, there are caveats.

Yes, we learned a lot.

Yes, we plan to do this again.

KRO + SDC Caveats

YANG has ordered Lists, YAML doesn't
https://github.com/sdcio/data-server/issues/394

NETCONF Actions are not declarative
https://github.com/sdcio/config-server/issues/347

KRO still early days and rather limited for full blown high level to low level config modelling

Benefits of GitOps with KRM

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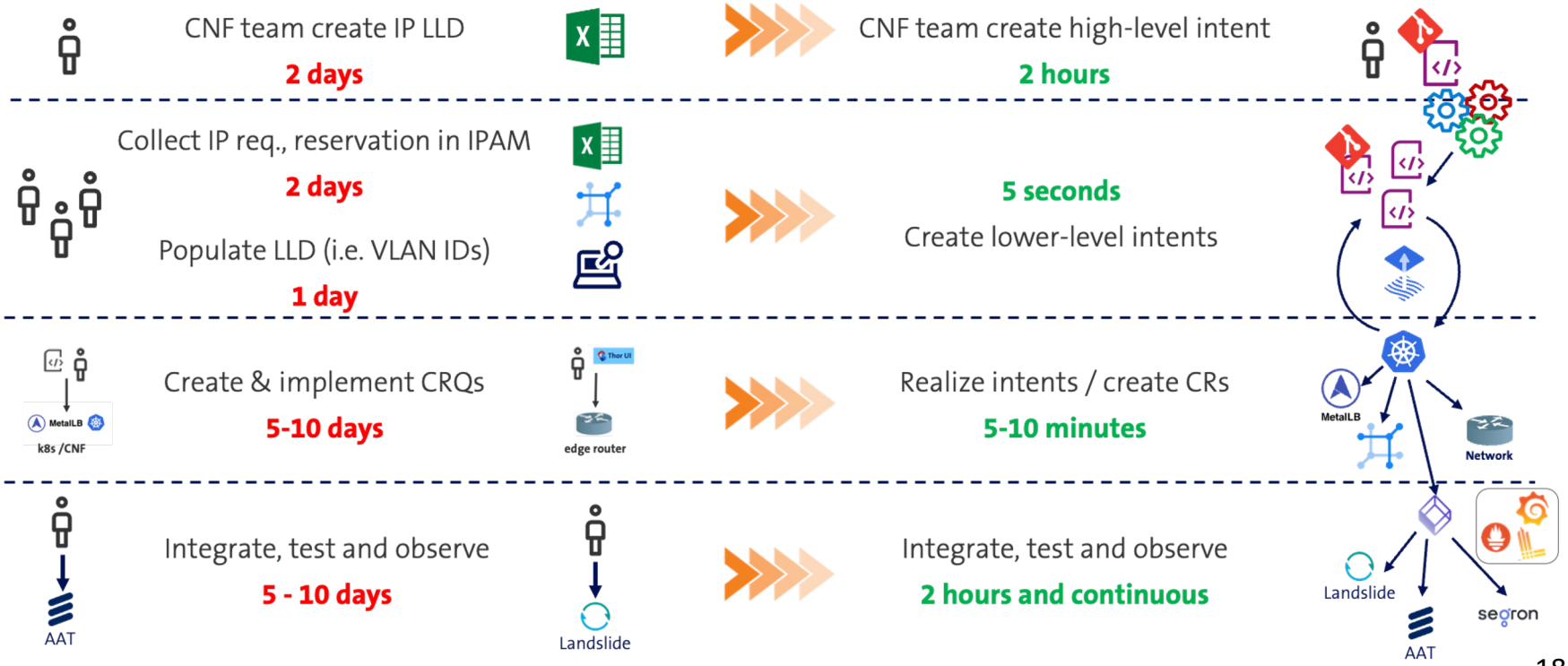
	Before	Now
Audit Trail	❌limited or unavailable	✅Everything tracked in git
Rollback Capability	❌manual	✅Built In
Approval Workflows	⌛CAB	✅Code Reviews
Testability	❌limited or unavailable	✅Automated testing
Consistency	❌limited	✅Git as Source of Truth
Scalability	❌limited	✅more of the same

Benefits of GitOps with KRM

Telco		TechCo
CNF team create IP LLD **2 days**	→	CNF team create high-level intent **2 hours**
Collect IP req., reservation in IPAM **2 days** Populate LLD (i.e. VLAN IDs) **1 day**	→	**5 seconds** Create lower-level intents
Create & implement CRQs **5-10 days**	→	Realize intents / create CRs **5-10 minutes**
Integrate, test and observe **5 - 10 days**	→	Integrate, test and observe **2 hours and continuous**



Challenges of GitOps + KRM in Telco

State Management: Telecom systems often have complex stateful components (databases, call routing) that don't necessarily fit well with GitOps

Configuration not GitOps native: NETCONF+YANG is not fully declarative

Skill Gaps: Operations teams traditionally focused on CLI/GUI tools; they need DevOps/SRE mindset

Process Redesign: Existing change advisory boards (CABs) and change windows may need restructuring

Working together to move forward

- From isolated to integrated staging & testing environments
- Joint Hackathons
- Working Group to provide future-proof and OCI compliant software delivery
- Working Group to shape the future of GitOps compatible artifacts
- Working together in the industry: Cloud Native Telco Forum (CNTF)

Join us! We need to evolve and learn together!

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CNTF

Inter-Vendor and Inter-Operator Forum to accelerate the Transformation Journey

github.com/swisscom/cntf

Kubenet

Helps network engineers understand the potential of Kubernetes for network automation

learn.kubenet.dev

Linktree

A collection of Cloud Native Telco Projects

linktr.ee/cloudnativetelco

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Feedback & Slides

https://sched.co/2DY39

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Joel Studler

Lead DevOps Engineer

Swisscom

Peter Woerndle

Senior Expert Deployment Architectures

Ericsson